Gradient-Enhanced Gaussian Process Regression for Non-Intrusive POD-Based Reduced-Order Models: An Adjoint-Informed 1D Diffusion Study

Andrew N. Jeffery [1,*], Jean C. Ragusa[1], Jim E. Morel[1]

[1]Department of Nuclear Engineering, Texas A&M University, College Station, TX

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ABSTRACT

Abstract:

- Expensive coupled-physics solves → cheap surrogate. This is the 1D sandbox version of that problem.
- Pipeline: FD diffusion solve → masked QoI → POD basis → GP surrogate for the POD coefficients.
- New idea: an adjoint solve gives free gradients of each POD coefficient. Train a gradient-enhanced GP (GE-GPR) on those instead of a value-only GP.
- Test cases: layered material properties, a smooth void/Teflon interface, maybe both combined.
- Punchline: GE-GPR hits the same accuracy with [???]× fewer training solves.
- Takeaway: adjoint gradients are a cheap way to make ROM surrogates more sample-efficient.

Keywords: reduced-order modeling, proper orthogonal decomposition, Gaussian process regression, adjoint sensitivity, gradient-enhanced surrogate

1. INTRODUCTION

- Expensive coupled-physics solves are too slow to call over and over (design loops, UQ). Non-intrusive ROMs fix that with a cheap parameter → QoI surrogate.
- Usual recipe: POD on a QoI snapshot matrix, project to a few coefficients $\alpha(\mu)$, fit a GP to $\mu \mapsto \alpha(\mu)$.
- Problem: a plain GP only learns from function values. Every expensive solve gives one data point per mode.
- Adjoints fix that almost for free: one adjoint solve per POD mode gives the exact gradient w.r.t. every parameter at once.
- So: feed those gradients into the GP too → gradient-enhanced GPR (GE-GPR / GEK). Cite [1, 2, 3, 4, 5].
- What this paper does:
  - derives the adjoint sensitivities of the POD coefficients for the diffusion solver (material layers, plus a smooth moving-interface geometry)
  - builds GE-GPR from scratch (scikit-learn has no gradient-observation support) and checks it against standard GPR
  - runs the test problems and how gradients impact the learning
  - lays out a verification (identity check, FD gradient check, sweep plots) that's reusable elsewhere.

*andrewnj03@tamu.edu

2. SANDBOX FORWARD MODEL AND BASELINE ROM

Background: The standard, gradient-free pipeline both test cases build on.

- 1D one-group diffusion, piecewise-constant material layers, Dirichlet BCs. Cell-centered 2nd-order FD $\rightarrow$ sparse system $A(\mu)\phi(\mu) = s(\mu)$
- Parameters μ: per-layer D, Σ_a, q
- QoI = a masked window of the flux, $y(\mu) = H\phi(\mu)$. Snapshot matrix $\rightarrow$ thin SVD $\rightarrow$ rank-r POD basis Ψ by an energy cutoff
- Coefficients by projection: $\alpha(\mu) = \Psi^\top y(\mu)$
- Baseline surrogate: one GP per mode, ARD-RBF kernel (scikit-learn)
- Error metrics used everywhere after this: field error ε_y, compared against the POD floor $\varepsilon_{\text{proj}}$

3. ADJOINT SENSITIVITIES OF THE POD COEFFICIENTS

The gradients that feed Section 4.

- Standard trick: a scalar QoI $J = g^\top\phi$ comes from one adjoint solve $A^\top\lambda = g$, via $J = \lambda^\top s$
- Each POD coefficient α_k is exactly this kind of scalar QoI ($g_k = H^\top\psi_k$) $\rightarrow$ adjoint solve per mode gives $\partial\alpha_k/\partial\mu_j$ for every μ_j at once
- Material-layer case: clean closed forms for $\partial\alpha_k/\partial q_m$ and $\partial\alpha_k/\partial\Sigma_{a,m}$; $\partial\alpha_k/\partial D_m$ requires more work
- Geometry case: a hard material boundary isn't differentiable at the boundary. Fix: swap it for a smooth sigmoid switch at the one moving interface (void/Teflon), derive the chain rule through it (two knobs: interface location, transition sharpness)
- Same adjoint machinery works for both cases; geometry just swaps a 0/1 layer indicator for a smooth per-cell direction vector
- Verification, same three checks each time: identity check ($\lambda^\top s$ vs. direct projection, matches to solver tolerance), finite-difference gradient check, and a 1D sweep with a tangent-line overlay

4. GRADIENT-ENHANCED GAUSSIAN PROCESS REGRESSION

The main method contribution.

- Idea: train one GP per mode on both function values and adjoint gradients, not just values. A derivative of a GP is still a GP, so this is one joint covariance model.
- Needs an augmented kernel: value-value, value-gradient, gradient-gradient covariances (closed form for ARD-RBF). Bigger matrix ($M(1 + p_{\text{act}})$ instead of M), fit by maximizing the augmented marginal likelihood.
- Built from scratch, since scikit-learn's GP can't take derivative observations.
- Everything else about the surrogate matches the baseline (one GP per mode, same kernel family). Isolates the effect of adding gradients.
- Same code reused unmodified on the geometry case, just with a different sampler/adjoint front end feeding it. Might point to this demonstrating that the method doesn't care what μ physically means.

5. RESULTS

- **Material-layer case:** SVD spectrum/rank choice, projection-error floor vs. r, then a learning curve comparing GE-GPR vs. standard GPR at matched training-set sizes. Report as "GE-GPR at M=· matches standard GPR at M=·" [???].
- **Geometry case:** Run once with only the interface location active (close to a 1D problem with nearly

all the SVD energy in mode 1), and once with both parameters active. Compare GE-GPR's benefit in each.
- **Combined case [???]:** Material layers + geometry together, if there's room/time. Same comparison, higher active-parameter count.
- **Cost accounting:** An adjoint solve reuses the same factorized matrix as the forward solve, so it's cheap per sample; the trade-off is a bigger, pricier-to-fit kernel matrix.

6. DISCUSSION

- Gradient enhancement helps most when: parameter count is small, the forward solve is the expensive part.
- Helps least when: the coefficient map is already easy to learn from values alone (the simple geometry case)
- Known limitation: the augmented kernel grows with $M(1 + p_{\text{act}})$. Doesn't scale to large p as-is.

7. CONCLUSIONS AND FUTURE WORK

- Restate the contributions from Sec. 1, past tense, with real numbers filled in
- Future work: Pushing past $p_{\text{act}} \in \{1, 2\}$, maybe using the adjoint gradients themselves as a sensitivity/dimension-reduction screen; running the same recipe on the real coupled-physics stack; a multi-output version instead of one GP per mode.

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APPENDIX A. AUGMENTED KERNEL DERIVATIVE IDENTITIES

Full algebra behind the closed-form K_{fd}, K_{dd} terms in Sec. 4, for anyone re-implementing `GEGaussianProcess` from scratch.

APPENDIX B. VERIFICATION FIGURES

Overflow plot dump if space runs out in the main body: identity checks, FD gradient checks, for both test cases.